Wave Packets and the Fourier Transform

PHYS 2203 — Introductory Physics III

October 28, 2026

Overview

Last lecture introduced the free-particle Schrödinger equation and its simplest solution, the plane wave $Ae^{i(kx-\omega t)}$. A plane wave has a definite momentum, but the particle is equally likely to be anywhere. A real, localized particle must be described by a wave confined to a region of space. This lecture shows how to build such a **wave packet** by adding plane waves of different wave numbers. We start with periodic functions (Fourier series) and beats, then build a single isolated pulse (the Fourier transform). We close by finding the Fourier transform of a Gaussian wave packet, which turns out to be another Gaussian.

1 Fourier Series

Fourier series: any well-behaved *periodic* function of period λ_1 can be written as a sum of sines and cosines whose wavelengths are $\lambda_1, \lambda_1/2, \lambda_1/3, \ldots$ With $k_1 = 2\pi/\lambda_1$, the wave numbers are whole-number multiples of the fundamental, $k_n = nk_1$:

$$\boxed{f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(nk_1 x) + b_n \sin(nk_1 x)\right]} \tag{1}$$

The constant a_0 is the average value of f. If f is odd, $f(-x) = -f(x)$, only the sine terms appear. If it is even, only the cosines and a_0 appear. Using $e^{i\theta} = \cos\theta + i\sin\theta$, the same series can be written more compactly as

$$f(x) = \sum_{n=-\infty}^{\infty} A_n \, e^{ink_1 x}, \tag{2}$$

the form that leads to the Fourier transform below. The list of amplitudes is the function's **spectral content**. A musical note works the same way: a guitar string vibrates at its fundamental plus harmonics of various amplitudes.

Example: a square wave. A square wave of amplitude 1 and wave number k_1, centered so that it is an odd function, contains only sine terms, and only the odd harmonics:

$$f(x) = \frac{4}{\pi}\left[\sin(k_1 x) + \frac{1}{3}\sin(3k_1 x) + \frac{1}{5}\sin(5k_1 x) + \cdots\right]. \tag{3}$$

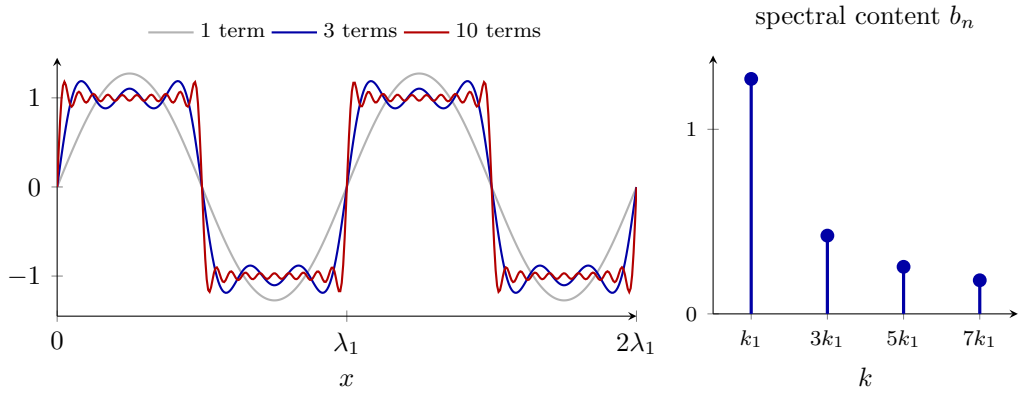


Figure 1: Left: adding more harmonics brings the sum closer to a square wave. Right: the amplitudes of the harmonics.

A sum of waves whose wave numbers are all multiples of k_1 always repeats with period λ_1. To describe a single particle we need a wave that does *not* repeat.

2 Two Waves: Beats

The simplest way to start localizing a wave is to add two waves of nearly equal wave number, k_1 and k_2:

$$y(x) = A\cos(k_1 x) + A\cos(k_2 x) = 2A\cos\left(\frac{\Delta k}{2}\,x\right)\cos(k_{\text{avg}}\,x)\,, \tag{4}$$

where $\Delta k = k_2 - k_1$ and $k_{\text{avg}} = (k_1 + k_2)/2$. This uses $\cos\alpha + \cos\beta = 2\cos\frac{\alpha-\beta}{2}\cos\frac{\alpha+\beta}{2}$.

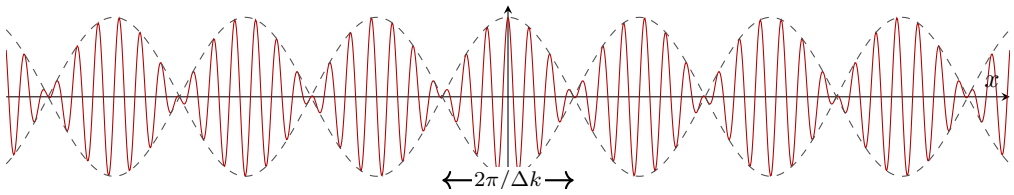


Figure 2: Two waves of slightly different wave number. The fast oscillation at k_{avg} is modulated by a slow envelope (dashed), forming groups of length $2\pi/\Delta k$.

The waves add constructively in some places and cancel in others, forming groups of length $\Delta x = 2\pi/\Delta k$. The closer together the two wave numbers, the longer the groups:

$$\Delta x\,\Delta k = 2\pi. \tag{5}$$

With only two waves the groups still repeat forever. Adding more waves with a range of wave numbers Δk makes the groups more isolated.

3 Nonperiodic Functions: The Fourier Transform

Figure 3 shows what happens as more waves are added, with wave numbers spread around k_0 and more closely spaced. In (b)–(d) the spacing between wave numbers is 1, $\frac{1}{2}$, and $\frac{1}{4}$ of a unit, so

the pulses repeat every 2π, 4π, and 8π units of x. The "impostor" pulses move farther away and disappear completely only when the wave numbers fill a *continuous* range (e).

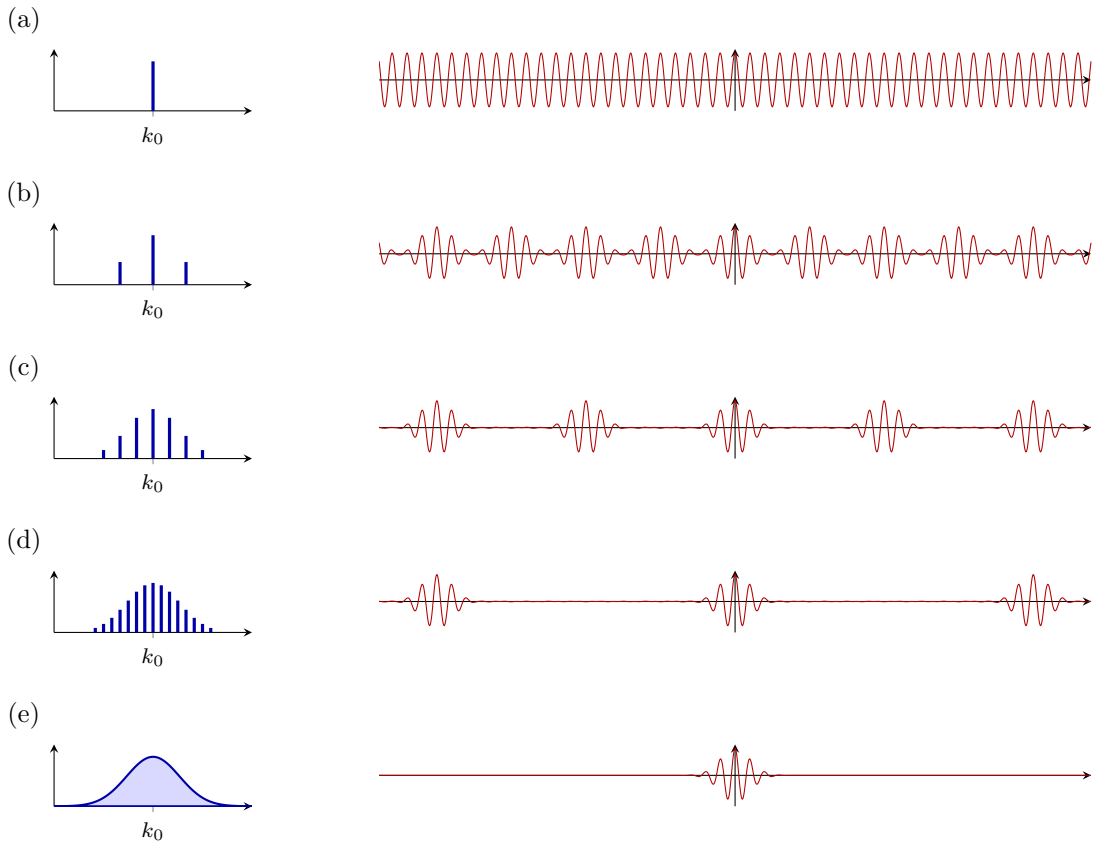


Figure 3: Building a single pulse from sine waves. Left: the amplitude of each wave number. Right: the sum. As the wave numbers become more closely spaced, the repeated pulses move farther apart. A single isolated pulse (e) requires a continuous range of wave numbers.

Fourier transform: any well-behaved *nonperiodic* function can be written as a sum of plane waves over a continuous range of wave numbers. Using the plane-wave building block $e^{ikx} = \cos kx + i \sin kx$ (at $t = 0$),

Fourier transform.

$$\psi(x) = \int_{-\infty}^{+\infty} A(k)\, e^{ikx}\, \mathrm{d}k, \qquad A(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \psi(x)\, e^{-ikx}\, \mathrm{d}x.$$

$A(k)$ is the **Fourier transform** of $\psi(x)$. It is the amplitude of the plane wave with wave number k, the continuous version of the A_n of a Fourier series. The second formula lets us find the amplitudes needed to build any given $\psi(x)$. ($\psi(x)$ is the wave function at a single time. We write $\Psi(x, t)$ when time is included.)

4 The Gaussian Wave Packet

A good model of a localized particle is the **Gaussian wave packet**

$$\boxed{\psi(x) = C\, e^{-(x/2\varepsilon)^2}\, e^{ik_0 x}} \tag{6}$$

- $e^{-(x/2\varepsilon)^2}$ is a bell-shaped "Gaussian bump" centered at $x = 0$. Its width is set by ε.
- $e^{ik_0 x}$ is a plane wave with wavelength $\lambda = 2\pi/k_0$.
- C sets the height.

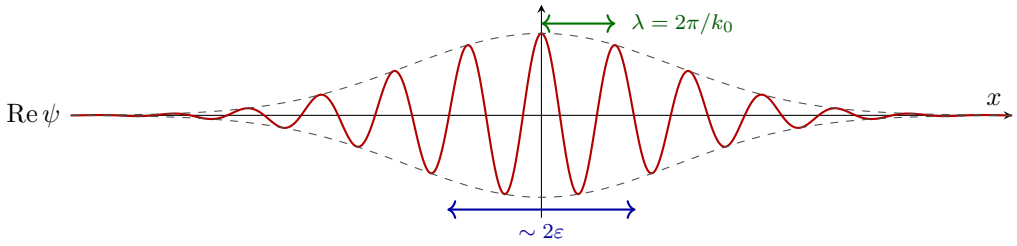


Figure 4: The real part of a Gaussian wave packet: a plane wave of wavelength $2\pi/k_0$ inside a Gaussian envelope (dashed) of width set by ε.

Its Fourier transform.

$$A(k) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} C\, e^{-(x/2\varepsilon)^2} e^{ik_0 x} e^{-ikx}\,\mathrm{d}x = \frac{C}{2\pi}\int_{-\infty}^{+\infty} e^{-x^2/4\varepsilon^2 + i(k_0-k)x}\,\mathrm{d}x. \tag{7}$$

This is a standard **Gaussian integral**:

$$\int_{-\infty}^{+\infty} e^{-az^2+bz}\,\mathrm{d}z = \sqrt{\frac{\pi}{a}}\, e^{b^2/4a}, \qquad \text{here } a = \frac{1}{4\varepsilon^2},\ b = i(k_0 - k). \tag{8}$$

Then $\sqrt{\pi/a} = 2\varepsilon\sqrt{\pi}$ and $b^2/4a = -\varepsilon^2(k-k_0)^2$, so

$$\boxed{A(k) = \frac{C\varepsilon}{\sqrt{\pi}}\, e^{-\varepsilon^2(k-k_0)^2}} \tag{9}$$

The Fourier transform of a Gaussian is another Gaussian, centered at k_0. To build the packet we need mostly wave numbers near k_0, with smaller amounts of the others.